Operator decision on off-shell competitors: the primary admissible class is

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The H-LAYER analytic closure (T-016 to T-021) holds on the registered admissible class; the adversarial review (T-023 v1.1) showed the sharp pin $T' \leq 13$ is class-specific and exhibited a lattice-but-off-shell escape ($N = 40$, $T' = 20$). This note records the operator decision on whether such off-shell competitors are physically admissible, with its physical justification. Scope: the Reading-H selection at the operating endpoint $I = 2 \times 10^{-3}$.

2. The decision

Off-shell / arbitrary-multi-radius antipodal lattice subsets are NOT included in the primary admissible class. The Reading-H comparison is a shell-selected Brazovskii comparison: physically relevant Bragg competitors live on the selected soft crystallographic shell or on registered shell-union degeneracies. The primary class is

$$\mathcal{A}_{\mathrm{adm}} = \big\{\text{registered crystalline shell / shell-union competitors}\big\}, \qquad T'(Q) \leq 13.$$

To make the proof adversarially safe, an extended fallback class is also retained:

$$\mathcal{A}_{\mathrm{ext}} = \big\{\, Q = -Q,\ \#Q \leq n_{\mathrm{pack}} \,\big\}, \qquad T'(Q) \leq \frac{N}{2} \leq 20.$$

3. Physical justification: the kinetic penalty

The Gaussian-Hartree kinetic kernel is

$$K_0(k) = r_R + c\,(|k|^2 - q_0^2)^2,$$

minimised on the soft shell $|k| = q_0$ with $K_0(q_0) = r_R = 0.3045$. An off-shell mode at the nearest distinct lattice radius $|k| = \sqrt{2}\,q_0$ carries

$$K_0(\sqrt{2}\,q_0) - r_R = 0.214 \text{ per mode } = \times 50 \text{ the selection MARGIN } (0.00432);$$

further radii cost more ($\times 3.8$, $\times 7.3$ in K_0 at $\sqrt{3}q_0$, $2q_0$). A competitor carrying off-shell modes therefore cannot be a LEADING Reading-H competitor: its kinetic cost exceeds the entire selection margin by $\sim 50\times$ per off-shell mode. Off-shell spectral content is Gaussian-sea / higher-energy / adversarial perturbation, not an independent Bragg competitor. Hence shell-supported is the primary physical class.

4. Two-tier closure

Primary $\mathcal{A}_{\rm adm}$	$T' \le 13$	$R_{\rm lead}(13) = 0.650$, joint $= \times 1.097$ (comfortable)
Adversarial $\mathcal{A}_{\rm ext}$	$T' \le N/2 \le 20$	$R_{\rm lead}(20) = 0.974$, joint $= \times 1.082$ (razor-thin)

On the primary class the H-LAYER analytic closure is comfortable. On the adversarial class the thresholds still hold ($20 < 20.5 < 26.2 < 60.4$), so off-shell escapes cannot break the closure – they only thin the RES-1 margin to $\times 1.026$. The proof is thus physically primary (shell-supported) AND adversarially safe (fallback-controlled).

5. Status

Operator decision recorded. The H-LAYER competitor-class sign-off is resolved: $\mathcal{A}_{\rm adm}$ (shell-supported) is the primary physical class; $\mathcal{A}_{\rm ext}$ (arbitrary antipodal) is the adversarial fallback. With this, the H-LAYER analytic residuals are closed on the primary class (comfortably) and on the adversarial class (thinly). B1/B2 may be re-examined for a hypothesis-reduced T6 (H-LAYER $\rightarrow$ shell-supported-crystalline completeness + inherited thin-certified grades); no tier flip is enacted here. B1/B2 T6 on {H-LAYER}, B5 T5.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"Excluding off-shell is just an assumption to dodge the escape."* DISMISSED: it is physically justified, not assumed – the K_0 kinetic penalty ($\times 50$ MARGIN per off-shell mode) makes off-shell competitors energetically uncompetitive at leading order. And the escape is NOT dodged: it is retained in $\mathcal{A}_{\rm ext}$ and controlled by $T' \le N/2 \le 20$.

(β) *"The adversarial fallback is razor-thin (RES-1 $\times 1.026$)."* ACKNOWLEDGED: admitting off-shell escapes thins RES-1 to $\times 1.026$. This is why the primary class is shell-supported (comfortable $\times 1.54$); the thin fallback is a reviewer-defence layer, not the physical operating point.

(γ) *"Does this flip B1/B2?"* DISMISSED: no. It resolves the competitor-class sign-off and ceilings any re-tier at a hypothesis-reduced T6. B1/B2 stay T6.

Quantitative sanity check (mandatory): *penalty* – $K_0(\sqrt{2}q_0) - r_R = 0.214 = \times 50$ MARGIN; *primary* – $R_{\rm lead}(13) = 0.650$, joint $\times 1.097$; *adversarial* – $R_{\rm lead}(20) = 0.974 < 1$, joint $\times 1.082$, $20 < 20.5 < 26.2 < 60.4$.

7. Result footer

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Result ID:              B5-BEYOND-LAYER-BOUND / operator decision on off-shell competitors
Precise statement:      DECISION: off-shell/arbitrary-multi-radius antipodal lattice subsets
                        are NOT in the primary A_adm (registered shell/shell-union, T'<=13),
                        justified by the kinetic penalty K_0(k)=r_R+c(|k|^2-q0^2)^2: an
                        off-shell mode at sqrt(2)q0 costs ~x50 MARGIN, so cannot be a leading
                        Reading-H competitor. They ARE retained in the adversarial A_ext
                        (Q=-Q, #Q<=n_pack), controlled by T'<=N/2<=20. Two-tier: A_adm
                        comfortable (R_lead(13)=0.650, joint x1.097); A_ext razor-thin
                        (R_lead(20)=0.974, joint x1.082); 20<20.5<26.2<60.4 survive.
Scope:                  Reading-H selection, operating endpoint I=2e-3. A_adm = primary
                        physical class; A_ext = adversarial fallback.
Dependencies:           aadm-exclusion-boundaries v1.1 (Boundary 3, fallback T'<=N/2);
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	hlayer-analytic-closure-consolidation v1.1; res5-chi-link-bypass (T'<=13); scscope-classwide-endpoint (joint, T'<60.4); Math424-AddA (K_0 kernel).
Evidence grade:	ANALYTIC (kinetic penalty + two-tier closure) + EXECUTED (res5_024_offshell_operator_decision.py 5/5) + OPERATOR DECISION.
Reproduction command:	python codes/vacuum/res5_024_offshell_operator_decision.py
Expected output:	off-shell penalty x50 MARGIN; A_adm R_lead(13)=0.650/joint x1.097; A_ext R_lead(20)=0.974/joint x1.082; 5/5 PASS.
Falsification gate:	An off-shell competitor whose total free energy (kinetic penalty included) beats the isotropic layer; or an A_ext set with T'>N/2; or a physical argument that off-shell modes are NOT kinetically penalised (would require a flat K_0 off the shell).
Tier before / after:	No tier flip. B1/B2 T6 on {H-LAYER}, B5 T5. The competitor-class sign-off is resolved: A_adm primary (shell-supported, comfortable), A_ext adversarial (fallback, thin).
No-overclaim statement:	Does NOT flip B1/B2. Records the operator two-class decision with its kinetic-penalty justification. The primary closure is comfortable (shell-supported); the adversarial fallback is razor-thin but unbroken. Any re-tier is ceilinged at hypothesis-reduced T6.
Next required action:	OPERATOR (optional): enact the B1/B2 re-tier to hypothesis-reduced T6 on the shell-supported class, or harden the inherited thin-certified grades (R_max, sunset, anchoring) for a stronger ceiling.